🌐 AWS Transfer Family – Explained

📌 What is AWS Transfer Family?

AWS Transfer Family is a **fully managed service** that allows you to **transfer files directly into and out of AWS storage** using traditional file transfer protocols:

- **SFTP** (SSH File Transfer Protocol)
- **FTPS** (FTP over SSL)
- **FTP** (standard File Transfer Protocol)

It enables **secure, scalable, and highly available file transfers** into **Amazon S3** or **Amazon EFS**—ideal for businesses that rely on **legacy systems or third-party partners**.

🧩 Key Features

Feature	Description
🔐 **Protocol Support**	SFTP, FTPS, FTP
☁️ **AWS Storage Integration**	Connects to **Amazon S3 or EFS** as backend
👥 **User Authentication**	IAM, custom identity provider (Lambda), or Active Directory
📈 **Scalable & Highly Available**	Managed infrastructure, built-in scaling
📊 **Logging**	Integration with **CloudWatch**, **CloudTrail**, **VPC Flow Logs**
🔄 **Real-Time Transfers**	No delay—files are available immediately in S3 or EFS
🛡️ **Security**	VPC support, IAM policies, IP whitelisting, encryption

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🔄 How It Works

[User's SFTP/FTP Client] ⇄ [AWS Transfer Family Endpoint] ⇄ [Amazon S3 or EFS]

- You create a **Transfer server** in AWS.
- Users connect using **SFTP/FTP/FTPS clients** (e.g., FileZilla).
- Files are stored directly in **S3** or **EFS**, based on your configuration.

📦 Use Cases

Use Case	Description
🔄 **Legacy Systems**	Continue using SFTP/FTP while migrating backend to AWS
🤝 **Partner File Exchange**	Partners upload/download files via familiar protocols
📁 **Data Lake Ingestion**	Ingest files into Amazon S3 for analytics
🧾 **Finance & Healthcare**	Compliant data exchange with audit logging
🎥 **Media Transfers**	Move large media files into cloud storage easily

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⚙️ Step-by-Step Setup Guide

✅ Scenario: Setup an SFTP server backed by Amazon S3

🔹 Step 1: Create an S3 Bucket (if not already)

1. Go to **S3 > Create bucket**
2. Name it (e.g., `my-sftp-bucket`)
3. Set permissions as needed

🔹 Step 2: Launch AWS Transfer Family Server

1. Go to **AWS Transfer Family > Create server**
2. Choose protocol: **SFTP**
3. Identity provider:
   - **Service-managed** (for demo/simple use)
   - **Custom Lambda** (for external users/DB)
   - **Directory Service** (for AD integration)
4. Endpoint type:
   - **Public** (for internet access)
   - **VPC-hosted** (private/internal access)
5. Choose logging options (CloudWatch)
6. Create the server

🔹 Step 3: Create a User

1. Go to the Transfer server → **Add user**
2. Configure:
   - Username
   - Role (IAM role with S3 access)
   - Home directory in S3 (e.g., `/my-sftp-bucket/user1`)
   - SSH public key (if using SFTP)

🛡️ IAM Role should include permissions like:

```
{
```

```
  "Effect": "Allow",
  "Action": [
    "s3:ListBucket",
    "s3:GetObject",
    "s3:PutObject"
  ],
  "Resource": [
    "arn:aws:s3:::my-sftp-bucket",
    "arn:aws:s3:::my-sftp-bucket/*"
  ]
}
```

🔹 Step 4: Connect Using an SFTP Client

- Use **FileZilla, WinSCP, or command-line SFTP**
- Host: The endpoint URL of your Transfer server
- Username: As configured
- Authentication: SSH private key or password (based on setup)

🔹 Step 5: Monitor Transfers

- Use **CloudWatch Logs** for session tracking
- Enable **CloudTrail** for API logging
- Use **VPC Flow Logs** if private endpoint

💰 Pricing Overview

Component	Price (Example)
Protocol Endpoint	~$0.30/hour per endpoint

Data Uploads to S3	Free
Data Downloads	Standard S3 rates (e.g., ~$0.09/GB)
EFS Usage	Charged separately
CloudWatch Logs	Based on log volume

📎 [AWS Pricing Page – Transfer Family]

🔐 Security & Compliance

- Supports **encryption in transit (TLS/SFTP)**
- Can restrict access via:
  - **IP allow lists**
  - **IAM permissions**
  - **Chroot-style S3 folder isolation**
- Compliant with **HIPAA**, **PCI-DSS**, **SOC**, and more

📝 Notes for Interview / Study

- AWS Transfer Family supports **file-level access via legacy protocols**
- Data is stored in **S3 or EFS**, not on a managed FTP server
- Integrates with **CloudWatch**, **IAM**, **VPC**
- You **don't need to manage servers or FTP software**
- Identity options:

- IAM users (simple)
- Custom Lambda (scalable/multi-tenant)
- Active Directory (corporate access)